PART A: Multiple choice or Short answer

(5 mark for each correct answer/choice)

Problem A1. Find the limit if it exists.

$\lim_{x\to-\infty}\left(\sqrt{x^2-2x+3}+x\right).$

Write your answer here:

Problem A2. Let $f(x)=e^x+2\sin x$, and $g(x)=x+1$ if $x>0$ and $g(x)=x^2+3x+1$ otherwise. If $h=g\circ f$, then $h'(0)=$?

Write your answer: $h'(0)=$

Problem A3. Evaluate

$\lim_{x\to-1}\frac{x^{2026}-1}{x+1}.$

Write your answer here:

Problem A4. Find $f'(0)$ if it exists:

$$f(x)=\begin{cases}x^4\sin\frac{1}{x} & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$$

Circle the best answer.

A. 1

B. 0

C. 2

D. It does not exist

Problem A5. Find the limit if it exists

$\lim_{x\to-1}\frac{|x-2|-|x+3|+x}{x+1}.$

Circle the best answer.

A. -1

B. $-\infty$

C. 1

D. A,B,C are incorrect

Problem A6. Find all vertical lines of the curve

$y=\frac{2x^2-8x-10}{(4x-1)(x+1)}$

Write your answer:

Problem A7. If $f'(x)\leq 6$ for all x and $f(2)=-1$ then the largest possible value of $f(8)$ is ..

Circle the best answer.

A. 35

B. 31

C. 50
D. A,B,C are incorrect

Problem A8. Find the 2026th derivative of $f(x) = \sin(3x)$.
Write your answer here:

PART B: WRITE YOUR FULL ANSWERS. *(If the space is not enough, please write on the back and indicate with a phrase like 'Problem ...continue on...')*

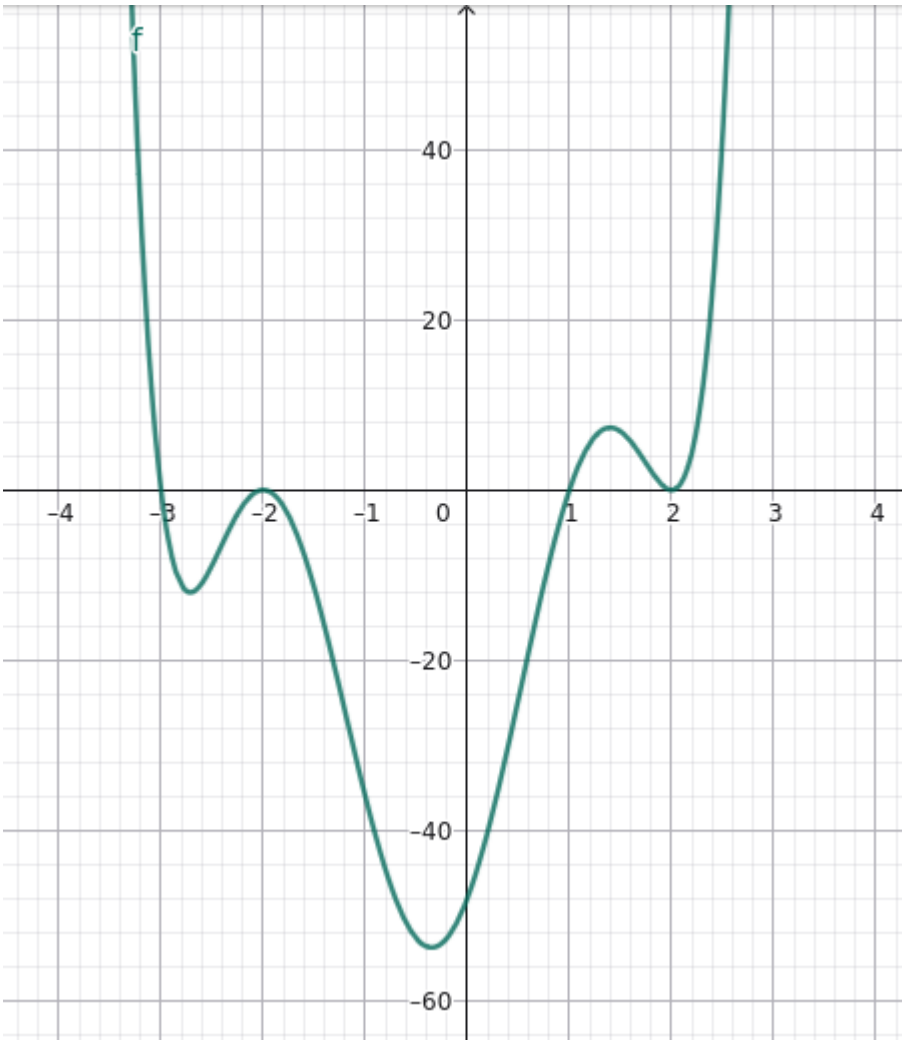
Problem B1. Find all values of a and b such that f is differential everywhere:

$$f(x) = \begin{cases} a\cos x + b & \text{if } x < 0 \\ x^2 - 2x - 3 & \text{if } x \geq 0 \end{cases}$$

Problem B2. Find equations of the tangent line at the point $(1, 1)$ to the curve $y = (\sin x)^{\ln x}$

Problem B3. Let $f(x) = x^3 - 3x^2 + mx - 1$, where m is a parameter (a real number).
a. Use the definition of derivative (Section 2.7, Definition 4 - Textbook) to find the derivative of f at the number a.
b. Show that the equation $f(x) = 0$ has at least one real root.
c. Find all m such that $f(x) = 0$ has exactly one real root.

Problem B4. Given the polynomial $f(x)$ with graph below (Assume all zeroes are real).
a. State $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$?
b. How many points where the tangent lines are horizontal? Estimate local minimal/maximal values of f?
c. Use the graph to name the zeroes of f, then state whether their multiplicity is even or odd
d. state the minimum possible degree of f and write one possible function for f in factored form;



Problem B5. Suppose f is a function with the property that $|f(x)| \leq x^4$. Find $f'(0)$?

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